



Women are too easily offended, but robots aren't: a feminist critique of sentiment analysis

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Received: 11 November 2024 / Accepted: 21 August 2025 / Published online: 2 September 2025
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Abstract

The history of the study of emotions has given way to several disciplines that have expanded our knowledge of the human mind, like psychology and philosophy. In our current times, Sentiment Analysis presents itself as a new path to study this field through automated and computerized means. However, automated systems can also carry many biases, such as sexism, thus possibly skewing their analysis towards a perpetuation of misogynistic views. By employing different Sentiment Analysis models on two sexism inventories, we observe that the emotional analysis these methods carry out varies in effectiveness and that this analysis can only reach a superficial understanding of sexism. Through these findings, we elaborate a critique of the technological concept of objectivity and neutrality in science. This study brings the feminist epistemological view to light to warn against an uncritical, indiscriminate use of technologies that can result in the further perpetuation and standardization of sexist biases in science.

Keywords Sentiment analysis · Epistemology · Feminism · Neosexism · Artificial intelligence

1 Introduction and object of study

The study of human emotions has been an area of academic fascination for centuries. From the many philosophical perspectives on people's inner lives to the development of scientific disciplines like psychology and neuroscience, many scholars have delved into the identification and explanation of those states that arise in an almost innately familiar way yet remain a mystery in many aspects. In the era of automation, as many fields become new experimental grounds for Machine Learning (ML) and Artificial Intelligence (AI), Computer Science has also taken its turn in the exploration of emotions through the development of Sentiment Analysis (SA) techniques.

The use of the expression “Women are too easily offended” in the title is a direct quote of the fifth item in the Ambivalent Sexism Inventory (the second item in the list of Hostile Sexism), which is employed in the study.

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The field of Sentiment Analysis has supposed a new step for the understanding of the relations between the human mind and computer science. SA, also known as Opinion Mining, is defined as “the field of study that analyzes people's opinions, sentiments, evaluations, appraisals, attitudes, and emotions towards entities such as products, services, organizations, individuals, issues, events, topics, and their attributes” (Liu 2012, p. 7) through Machine Learning and other automated methods, and consists in the identification of emotional content in textual data. To do this, Sentiment Analysis techniques work in three different levels. The first tier is known as the document level, which looks at the overall sentiment conveyed throughout the entire text being analyzed. At this level, it's assumed there's only one main subject of opinion, so it's not ideal for comparing different topics. Moving to the second tier, we have the sentence level, which focuses on opinions expressed in individual sentences. This helps break down the overall sentiment of the text and understand its details more clearly. Lastly, we have the aspect level, also known as the entity or feature level. Here, the analysis zooms in on how different words and expressions are perceived, allowing us to grasp the importance of various linguistic mechanisms like negation in sentences and similar language patterns. Many systems analyze the aspect level first and add up the emotional content found

to generate the sentence level analysis, doing the same again with the results from this level to achieve the final document level results (Medhat et al. 2014). These concepts found the base framework for SA models.

The results for the different levels are interpreted and expressed by the Sentiment Analysis system in terms of polarity, which was initially established as a binary variable that could be “positive” or “negative”. More recently, many Sentiment Analysis systems have started to evaluate the text within a polarity spectrum, which assigns gradual values between the extremes of “positive” and “negative” (Pang and Lee 2005). This has made Sentiment Analysis techniques very popular in fields where human interaction is of high relevance, like communications, where SA is employed for studying trends and opinions on specific events or topics (Drus and Khalid 2019; Yue et al. 2019), education, where it is mainly used to monitor students’ emotions during class (Mite-Baidal et al. 2018; Zhou and Ye 2023) or marketing, where SA models are used to study product reviews and give brands useful information for the development and marketing of their products (Rambocas and Pacheco 2018). This proliferation of Sentiment Analysis methodologies has also taken hold of the world of research, as it has started to be implemented more frequently in different areas of studies.

This field has seen very rapid development, mainly through the use of AI technologies and the introduction of ML and Deep Learning (DL) into the field. This field includes many different methodologies that can be implemented to extract the polarity of text. Within these methodologies, we can find lexicon-based approaches, which are based on dictionaries or corpuses, which are collections of words that the system can identify and evaluate, and which are associated with a certain polarity through training (Wankhade et al. 2022). However, newer techniques involve ML and DL techniques to improve the analysis and the use of contextual information. By employing neural networks, which are formed by layers that process the information they receive in and pass them onto the next one, SA has been developed into more complex models that go beyond associating a polarity to each word or expression (Rojas-Barahona 2016; Tang et al. 2015). The introduction of attention mechanisms and the Transformer architecture was a great step for SA, as the models developed from it can relate the words in a sentence to one another to extract contextual information and better evaluate the input (Vaswani et al. 2023). Overall, the field of SA keeps on developing at a quick speed, mainly in its technical aspects.

Innovation, however, is not the only relevant factor that comes with technological advancements, and while the use of state-of-the-art tools is highly encouraged in many workplaces and fields of study, the early adoption of these devices and methods can serve to perpetuate existing problems in research or lose control of our work. While Sentiment

Analysis offers an interesting and new methodology for many tasks and fields, a deeper understanding of what automated emotional analysis implies is necessary before truly introducing it into our understanding of the human mind.

2 Theoretical framework

2.1 Technological knowledge and social detachment

Technology surrounds us in every aspect of our lives in a deeper way than it has ever before in history: our information, work, resource management, communication and even our leisure are for the most part dependent on technological tools and implementations. As Adorno (1998) once put it, we are currently technological people. This has resulted in a generalized perspective on technology as a daily element of life through which the fascination and suspicion felt by earlier generations for new technical advancements, like the critique towards printed media after Johannes Gutenberg unveiled the first press in the XV century, has given way to a more superficial outlook on the implications that the tools we use have as representations of the social structures we live in. As such, technology becomes fetishized (in the critical philosophical sense of the word), and its creation and use are taken as individual acts mostly separated from our relations to other humans and to humanity as a collective.

In a society whose ruling structures have historically perpetuated a system of oppression that is sexist, racist and capitalist, a fetishized superficial view of technology results in a separation between technology and social conscience. Marginalized communities are put in danger of suffering through the use of technological tools, which are considered to be independent and neutral objects by this fetishized view (Joque 2022; Wacjman and Young 2023). Plant (2000) explains this fetishization and the view resulting from it as parallel to marriage vows, and in specific as a marriage of the machines to the system under which they are developed, thus serving it and working to maintain it. However, she is optimistic about the possibilities of liberation that these technologies bring forth to create networks that will help women’s struggle (1997), much like Haraway when she explored robotics and technology as a way to free society from gender identity and oppression (1991), thus showing the contradictory nature of technology and its fetishization in our current system.

The possibilities that technology offers for human activity then clash with the way society employs technological devices. In a similar vein to Adorno, Marshall McLuhan developed a theory of technology as an “extension of man” according to which technological advancements are made to fulfill a design intent or necessity but also carry a message

(a social implication) which does not appear immediately to individuals or society (McLuhan 1988). According to McLuhan, people employ and create technologies without being conscious of the social factors that their technology represents, factors that are influential both in the development and in the latter use of the tools. As technology develops further and gains more relevance in society, it becomes a crucial element in shaping people's behaviors too, as a lack of conscious thought on the structures behind technology solidify them and drive us into perpetuating them further (McLuhan 1962). This technological determinism allows for technological development to see itself mostly unchallenged by different social causes that struggle with existing oppressive structures, like feminism (although there are exceptions to this like some technofeminist currents), and thus these structures see themselves reinforced.

There is another effect of this on the users, due to the increasing complexity of the technological tools we employ and the lack of knowledge people tend to have about them. This lack of knowledge, which aids the fetishized view of technology to maintain itself as it dissuades people from looking deeper into the technologies they use (Cao et al. 2021), also results in feelings of powerlessness and alienation (Lasch 1979; Minch and Ray 1986), which isolate users and hold them back from involving themselves further in knowing these technologies. For communities that have already seen themselves historically isolated or disadvantaged, this can also imply that technologies are not made to accommodate their needs. These communities might experience disenfranchisement from access to technologies and might be culturally and economically gatekept from them, as in the case of many disabled people being barred from integrating into online spaces for a long time due to hardware and software design not taking them into account (Bowker 2010). This situation, in turn, results in a lack of engagement with these communities when it comes to designing new technologies.

2.2 Emotional hierarchies and the feminist epistemological view

In fact, the way emotions are conceived in the development of Sentiment Analysis technologies brings forth another concern that puts the adequacy of these systems in question. For most Sentiment Analysis models, emotions can be classified as being positive, neutral, or negative (Esuli and Sebastiani 2006; Pang and Lee 2005), with the positives mainly relating to happiness and satisfaction, the negatives with sadness and anger, and the neutral emotions being reduced to a sort of indifference or lack of significance. This categorization, harmless in its appearance, relates to specific political and cultural views of emotions and their social implications.

The positive aspects of an emotion are not absolute or natural, but rather an interpretation of how this emotion can help contribute to an individual's satisfaction of objectives that fall in line with the socially accepted paths a person may take in life (Ahmed 2014; Davies 2015), relating these emotions directly to a series of social imperatives that are seen as normative. In this way, emotions are hierarchized and "positive" emotions become elements to be cultivated and reinforced while "negative" emotions must be suppressed or repurposed, and techniques such as mindfulness lessons are developed that not only reaffirm this idea, but also help implement it further within the general population (Cullen 2011). In the sexist capitalist system we live in, this will mean that the way the emotional hierarchy is set up will, in some cases, serve as an obstacle for practices and discourse that position themselves in opposition to oppressive environments and situations, such as complaints and denouncement of sexist behaviors and actions (Ahmed 2021; Illouz 2008). This hierarchization ultimately serves to keep the inequality that is structurally enforced in society.

Part of the political hierarchization of emotions has also involved the subordination of these to reason as a more relevant aspect of decision making (Ahmed 2014), thus underestimating the effects emotions have on the way we direct ourselves in life (Gill 2017; Illouz 2007), which are more pervasive than it is usually mentioned. This has, historically, been tied to the association of womanhood with emotional chaos and inferiority, as expressed by Kant with his quote "it is difficult for me to believe that the fair sex is capable of principles" (Frierson 2011, p. 39). The misogyny present in the history of the social understanding of emotions can be present in SA systems too, as the data they learn from has been created in the same society that has held and currently holds the structures that allow for sexist behavior and thought. This could manifest by considering negative expressions directed against women as positive or neutral or by judging emotions that have historically been considered to be womanly as negative.

In neoliberal discourse, positive emotions are directed towards the accumulation of wealth and the conception of the self as an autonomous, independent individual (Arthington 2016), resulting in a state of alienation and competitive hostility towards others (Becker et al. 2021) that fosters emotional unrest. This relates to what Lauren Berlant (2007, p. 33) defines as cruel optimism, "a relation of attachment to compromised conditions of possibility whose realization is discovered either to be impossible, sheer fantasy, or too possible, and toxic". This definition is a way to question how the pursuit of positive emotions and goals affects people within the neoliberal context. This discomfort becomes more generalized as material conditions make it increasingly harder for most of the population to achieve the goals set by the capitalist standard in the first place (Palmer 2013), keeping people

in a painful situation of constant dissatisfaction. If the emotional classification of Sentiment Analysis systems reflects the same hierarchy as the one established by the “common sense” of capitalist structures, then these tools may serve to keep people tethered to harmful psychological patterns in the many different fields where they can be employed, and even to make it harder for individuals and collectives to find other paths that represent healthier outcomes.

In the way they present themselves, Sentiment Analysis systems seem to offer a new convenient way to study people’s emotions with a velocity and efficiency that has never been seen before, but they do not offer a view of the world they exist in. For decades, feminist epistemologies have denounced the way that science is considered to create “objective” knowledge while supporting the disenfranchisement of marginalized groups (Harding 1996; Wigginton and Lafrance 2019), and we can now point to the “objective” technologies that are born from this science as tools to enable this perpetuation of oppressive conditions (Fee 1981). The study of emotions and psychology in general have social, human and philosophical roots which we must not forget, and the overreliance on quantitative research (Bennett 2021) and the adoption of ways to automate it suppose a danger for qualitative inquiry and critical work.

While there have been studies that tried to adapt Sentiment Analysis models to a more feminist perspective (Abburi et al. 2024; Rodriguez et al. 2020) or SA implementations that attempt to tackle specific social issues, such as hate speech detectors (Jiang and Suzuki 2019), these tend to have a lower precision than more general models or be limited in scope. Furthermore, most research that attempts to introduce sexism detection into Sentiment Analysis does not take into account the epistemological questioning of these tools in the first place, and treats sexism mainly in a linguistic, explicit sense, which does not completely represent the structural and more subtle ways in which sexism, racism, or capitalist ideas are sustained.

2.3 Different modalities of sexism

In this study, we will focus on the subtle and varied forms of sexism that could represent a challenge for Sentiment Analysis models and the conception of emotions that is associated with them. Sexism has at its core the purpose and utility of maintaining men’s domination over women (Swim and Campbell 2003), but this purpose can be conveyed in different ways when it is expressed. The theory of ambivalent sexism divides these expressions into two categories, hostile sexism and benevolent sexism (Glick and Fiske 1996). Hostile sexism, which is also commonly referred to as classic misogyny, is the more explicit and violent representation of sexism through expressions, beliefs, or actions that follow the traditional conception

of prejudice and carry harmful intentions towards women. On the other hand, benevolent sexism upholds traditional and sexist stereotypes about women, but it is expressed in a way that is perceived as positive in tone and is related to caring, protection and helping women. This means that, from the point of view of the current hierarchy of emotions, expressions that carry benevolent sexism are mostly viewed as positive and harmless while they maintain the status quo. On the basis of this distinction, Glick and Fiske developed an Inventory of items to distinguish people’s acceptance of hostile and benevolent sexism.

Glick and Fiske’s inventory is, however, not the only sexism inventory that is currently studied in psychology. The Neosexism Inventory (Tougas et al. 1995) studies a specific expression of sexism that belongs to more recent times. In a context in which social progress has made it less acceptable to express traditional misogynist views, neosexism appears as a form of sexism that hides itself as an opposition to the political and social demands of feminism, maintaining dominance of men over women through the characterization of women’s progress as a threat to men. Both neosexism and benevolent sexism are ways in which sexist expressions hide themselves in a way that makes them less likely to be challenged, and thus might also be harder to detect for Sentiment Analysis models.

2.4 Aims and questions

This study aims to observe the relation between Sentiment Analysis models and different types of sexism in language as a way to examine how these technologies interpret sexist expressions and emotions in order to discern what role they can have in the perpetuation of the oppression of women. To do this, we will employ Sentiment Analysis using three different models on two sexism inventories to simulate the usual implementations of Sentiment Analysis models. The SA models employed were VADER, DistilBERT and RoBERTa. Given the main objective, different research questions have been elaborated.

- **RQ1** Do Sentiment Analysis models classify sexist sentences as negative, or do they consider them to be either positive or neutral?
- **RQ2** Are there differences in the evaluation that Sentiment Analysis models do of language that expresses Benevolent Sexism, Hostile Sexism or Neosexism?

Table 1 Sample sentences from the database alongside the inventory they come from

Sentence	Original inventory
Most women interpret innocent remarks or acts as being sexist	Ambivalent Sexism (Hostile sexism)
Women seek to gain power by getting control over men	Ambivalent Sexism (Hostile sexism)
Women, as compared to men, tend to have a more refined sense of culture and good taste	Ambivalent Sexism (Benevolent sexism)
A good woman should be set on a pedestal by her man	Ambivalent Sexism (Benevolent sexism)
Over the past few years, women have gotten more from government than they deserve	Neosexism
I consider the present employment system to be unfair to women	Neosexism

3 Methodology: quantitative analysis of SA models

3.1 Data collection and procedure

To carry out this study, the first step was to gather a database of sentences that could serve as an itemized list of different forms of sexist language. To do this, we employed the items that can be found in Glick and Fiske's Ambivalent Sexism Inventory (1996) and those given by Tougas et al. (1995) in their Neosexism Inventory. While these are not examples of collected utterances, they are written instances of sexist thought and behavior that have a great precedent in psychological research. All of the items were taken as they were written in the inventories and put in order in a datasheet that was then inputted into a Python notebook.

Once the sentences were all collected, they were put in an Excel sheet. After this step, a total of 33 sentences made up the database, out of which 22 came from the Ambivalent Sexism Inventory and 11 were taken from the Neosexism Inventory. This sample size was considered acceptable as the high precision of the already trained standard Sentiment Analysis models makes it sufficient to work with this amount of samples and due to the relevance of these two specific inventories in the field of sexism research, as they are the most employed sexism inventories among the available ones. These sentences were all classified manually in accordance with the different types of sexism that are included in this study, that is, they were categorized as either neosexism, hostile sexism or benevolent sexism. Given that 7 of the 33 sentences in the inventories are actually not sexist (with the other 26 being explicitly sexist) and are instead used to measure negative reaction to them (see Table 2), these were classified as non-sexist, but were put through the analysis process nonetheless as they too were considered to be of interest. Table 1 shows some of the sentences gathered for the database.

Once the sample database was completed, the data was pre-processed using Python to make it appropriate for analysis. For the VADER analyzer, the first procedure realized was lemmatization. This is a Natural Language Processing (NLP) technique employed to reduce words to their base or

root form, referred to as a lemma. It employs a systematic approach to normalize the text by transforming diverse word variations, encompassing plurals and various verb tenses, into their shared base form. This process helps to facilitate a more precise and efficient analysis of text data, as it enables the grouping together of words with similar meanings and reduces the computing load required by the system used. The lemmatizer employed was the WordNet lemmatizer from the NLTK library, which uses the WordNet dictionary to find the lemma of each word in the sentence (Bird and Loper 2004; Miller 1995).

After the lemmatization process, the sample dataset underwent tokenization to make the data more appropriate for the analysis and support entity recognition. Tokenization is a crucial preprocessing step that involves breaking down the text into discrete units or "tokens," such as words, phrases, or individual characters. The fundamental purpose of tokenization is to structure the text effectively, simplifying subsequent analysis. This procedure includes the removal of punctuation and non-word elements, segmentation of the text into distinct words or phrases, and the assignment of unique numerical values to each token. Through this breakdown into smaller components, tokenization makes it easier for the computer to identify textual units to analyze. For the VADER analyzer, the tokenizer used was the WhiteSpace tokenizer from the NLTK library, which tokenizes sequences at a word level by creating a token every time it finds a white space in the sentence. In the case of the BER-based models, each used a pre-trained version of its own baseline tokenizer. The DistilBERT tokenizer, which employs WordPiece subword tokenization, turns each word into tokens that it can recognize and relate to each other (Devlin et al. 2019). The RoBERTa tokenizer, however, uses byte-pair encoding, which works by substituting the most used pairs of bytes in a sequence for a new byte, thus compressing the sequence and allowing for better processing (Gage 1994).¹

¹ The code for the analysis and pre-processing can be found in the following GitHub repository: <https://github.com/dariodona/Sexism-in-Sentiment-Analysis-data>.

Once the data was correctly processed, the Sentiment Analysis was applied to it using the VADER (Valence Aware Dictionary and sEntiment Reasoner) model, the DistilBERT model (Sanh et al. 2020) and a version of the RoBERTa model (Liu et al. 2019; Loureiro et al. 2022) specifically trained for SA. This was done in order to compare the different models, as VADER is a lexicon-based analyzer whereas DistilBERT and RoBERTa are both based on the Transformer architecture (Vaswani et al. 2023). Developed by researchers at the Georgia Institute of Technology (Hutto and Gilbert 2014), VADER utilizes a lexicon-based approach specifically designed for the analysis of social media content, although its applications have since been broadened. This SA model has been considered one of the most accurate lexicon-based models, outperforming other popular ones like TextBlob (Bonta et al. 2019). Notably, a multilingual version of VADER, denoted as vader-multi, was utilized in this study and seamlessly integrated into Python. DistilBERT and RoBERTa both use the BERT (Bidirectional Encoder Representations from Transformers) architecture, which is based on the Transformer one. These models of Deep Learning transform words into tokens, which go through many layers that extract information from them and contextualize them with each other in order to understand which are more relevant to the text. This allows for analysis with contextual awareness.

VADER relies on a sentiment lexicon containing words and their associated sentiment scores, ranging from -4 to $+4$. This scale represents extreme negativity to extreme positivity, with 0 signifying neutrality. The lexicon incorporates rules and heuristics to address special cases like negation and emphasis. The sentiment determination process involves locating tokenized text, scoring each word based on the lexicon, and aggregating these scores to generate an overall sentiment score for the data. This sentiment score ranges from -1 to $+1$, with negativity, positivity, and neutrality indicated by negative, positive, and zero scores, respectively. Normalization of valence values is carried out to obtain scores within -1 to 1 range using a specific formula. In the way Sentiment Analysis models interpret emotions, positive refers to a desirable emotion, one which is good to have or express, whereas negative emotions are the opposite of this and neutral refers to indifference or non-emotional expressions. VADER's analysis is further nuanced by considering the intensity of sentiment, adjusting scores for intensifiers like "very" or "extremely," and it can handle emoticons, slang, and informal language prevalent on different media. The outcomes provided by the model present results as a compound score reflecting the numerical value of polarity and intensity of said polarity.

Both DistilBERT and RoBERTa are SA models developed from the original BERT architecture, which is based on the Transformer Deep Learning architecture. The Transformer

architecture has become an essential element in Natural Language Processing (NLP), as many current models of Generative AI and Sentiment Analysis models are built on Transformer-based architectures (Bashiri and Naderi 2024; Yadav 2024). This architecture is based on attention mechanisms and, more specifically, self-attention. Attention mechanisms allow for the weighting of the different tokens in the sentence according to how relevant they are, thus relating the different tokens between themselves and obtaining contextual information (Bahdanau et al. 2016). Self-attention allows for the processing of a sequence of tokens (like a sentence) to be focused on in order to better weigh the words and understand the context of the sentence in itself and in a larger context (Vaswani et al. 2023), thus better capturing many complex aspects of language like polysemy.

The BERT architecture applies the principle of self-attention bidirectionally, meaning it can process and contextualize the sentence forwards and backwards, obtaining better results and gathering more information (Devlin et al 2019). When a sentence is inputted into a BERT-based model, it is first tokenized and transformed into a vector so that the model can read it, and the sequence generated is processed through self-attention layers that weigh the tokens in order to better get the context and intention of the sentence. The tokens are compared to a vocabulary of tokens that the models have already trained into them, from which the polarity scores can be calculated using the weights the model has given the tokens. The use of the DistilBERT and RoBERTa models specifically is due to them being state-of-the-art models that represent improvements on the BERT architecture either through compressing the model to reduce computational and environmental consequences (in the case of DistilBERT) or a more extensive vocabulary to obtain better results (RoBERTa).

The process for each model was applied to each sentence separately through Python automation. The resulting scores were taken and put in a table alongside the corresponding sentence. Using Python, the polarity scores were translated into polarity tags named "pos" (positive), "neg" (negative) and "neu" (neutral) depending on their value (0 for neutral sentences, positive scores for positive sentences and negative scores for negative sentences). These were added to the table with the corresponding sample sentence. The number of tags for each polarity was counted and added up, then turned into a percentage with respect to the total number of samples.

4 Results: Is sexism positive or negative for SA models?

In this section, we will show the different results obtained through the analysis of the Sentiment Analysis model dividing them between the results from the analysis of all

Fig. 1 General sentence evaluation by polarity expressed as a percentage per model

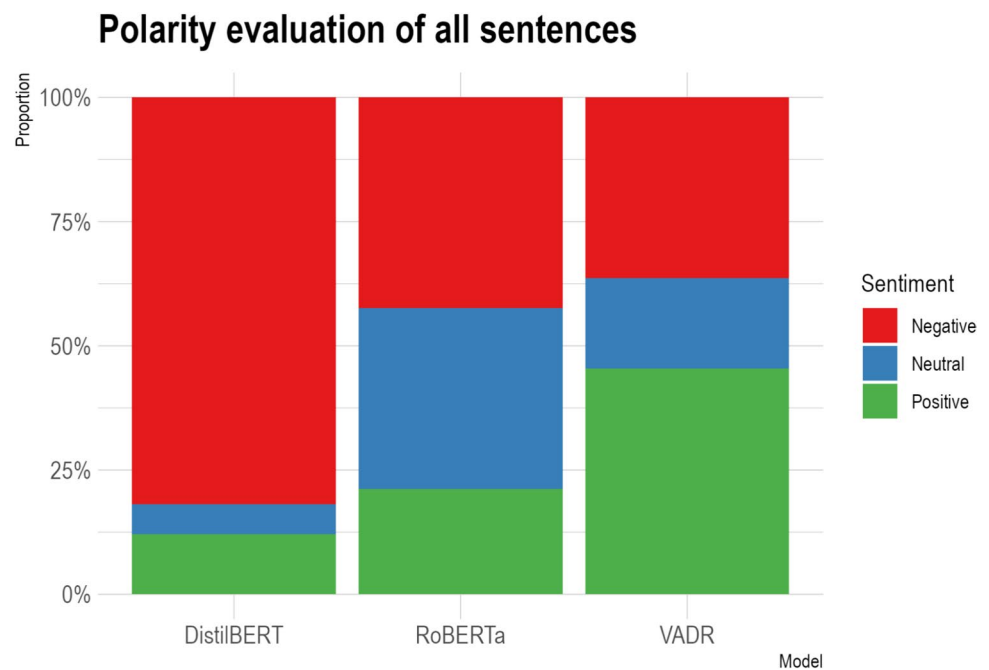
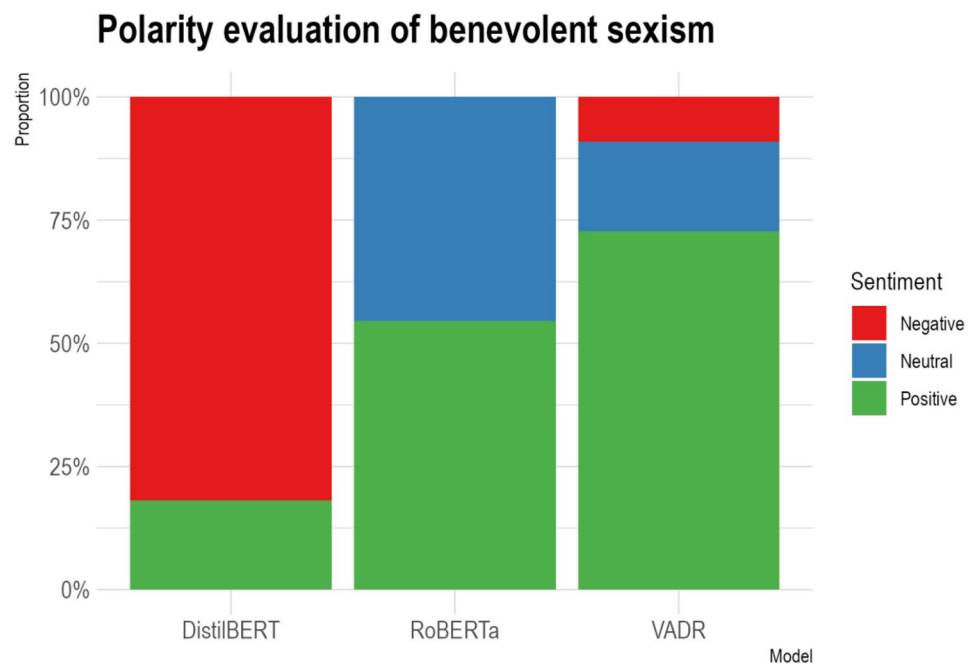


Fig. 2 Polarity evaluation of sentences that exemplify benevolent sexism per model



sentences in the database, the results when analyzing only the sexist sentences and the results that come from analyzing only the non-sexist sentences.

4.1 Analysis of the total sample of sentences

After putting the sample through the Sentiment Analysis models, all 33 sentences were evaluated by this model as either positive (that is, emotionally desirable), negative or

neutral. Figure 1 shows the general distribution of evaluation among the sentences.

As it can be seen in the figure, the models give different polarity results for the database. While DistilBERT identified over 75% of sentences as negative and less than 15% as positive, RoBERTa only evaluated less than 50% of the sentences as negative, giving a similar percentage of neutral ones. Finally, VADER classified almost 50% of the sentences as positive, showing less awareness to

sexism, while DistilBERT appears to be the most aware model of the three.

Diving into more specific categorizations, Figs. 2 and 3 represent the amount of sentences in the list of sentences associated with benevolent and hostile sexism, respectively, and the evaluation that the Sentiment Analysis model gave of them.

These figures show a stark contrast between the analysis of benevolent and hostile sexism. When it comes to Benevolent sexism, DistilBERT was the only model that classified more than 75% of the sentences as negative, while RoBERTa considered there were no negative sentences and VADER had the highest percentage of sentences considered positive, at almost 75%. This shows that, out of these models, DistilBERT is the only one that seems to be aware of benevolent sexism. On the other hand, the analysis of hostile sexism showed that the two BERT-based models classified over 50% of the sentences as negative while VADER's percentage of negative sentences was under 50% but, more significantly, was the same as the percentage of sentences it considered to be positive.

With regard to the Neosexism Inventory, Fig. 4 shows the way that the Sentiment Analysis model evaluated the sentences included in that classification. As it happened in Figs. 2 and 3, positive and neutral results have been added up to put in perspective the opposition of these two polarities and the negative one.

The analysis of the sentences belonging to the Neosexism Inventory shows that all models considered over 50% of the sentences to be negative, with DistilBERT being the model that classified the highest percentage of sentences as

negative, and VADER the one that had the lowest percentage. Both BERT-based models appear once again to be more aware of sexism than VADER, with DistilBERT being the most aware one for all types of sexism studied.

4.2 Analysis of the non-sexist sentences

As part of the general database, the 7 non-sexist sentences were also put through the Sentiment Analysis model, which means they too have a polarity associated. To analyze them separately and study the way Sentiment Analysis models evaluate non-sexist sentences as opposed to sexist ones, Table 2 shows the sentences that were not considered to be sexist in the inventories as well as the score given to them by the Sentiment Analysis model, their polarity expressed as “pos”, “neu” or “neg” (Fig. 5).

Out of all the models, DistilBERT was the one to classify the most non-sexist sentences as positive, while having the same percentage of negative sentences as VADER. RoBERTa, however, classified over 50% of the sentences as neutral and had the least negative sentences. This lines up with RoBERTa being the model with the most neutral sentences and might point to a lower polarization of gender-related sentences for it.

4.3 Analysis of the explicitly sexist sentences

Removing the 7 non-sexist sentences, a total of 26 explicitly sexist sentences were evaluated. Figure 6 shows the general polarity evaluation of these 26 sexist sentences included in the list.

Fig. 3 Polarity evaluation of sentences that exemplify hostile sexism per model

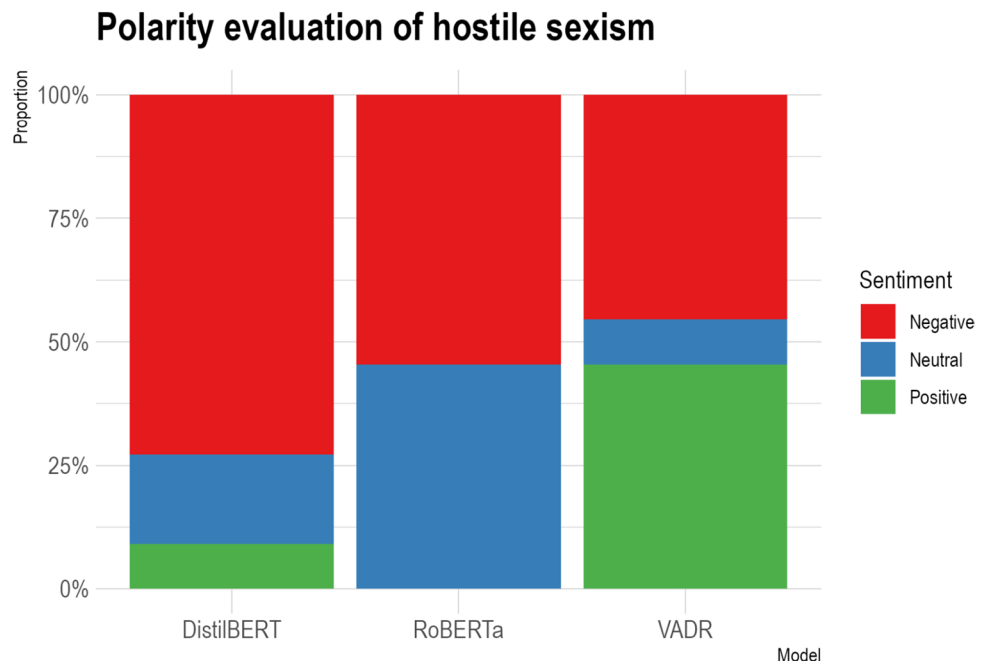


Fig. 4 Polarity evaluation of sentences that exemplify neo-sexism per model

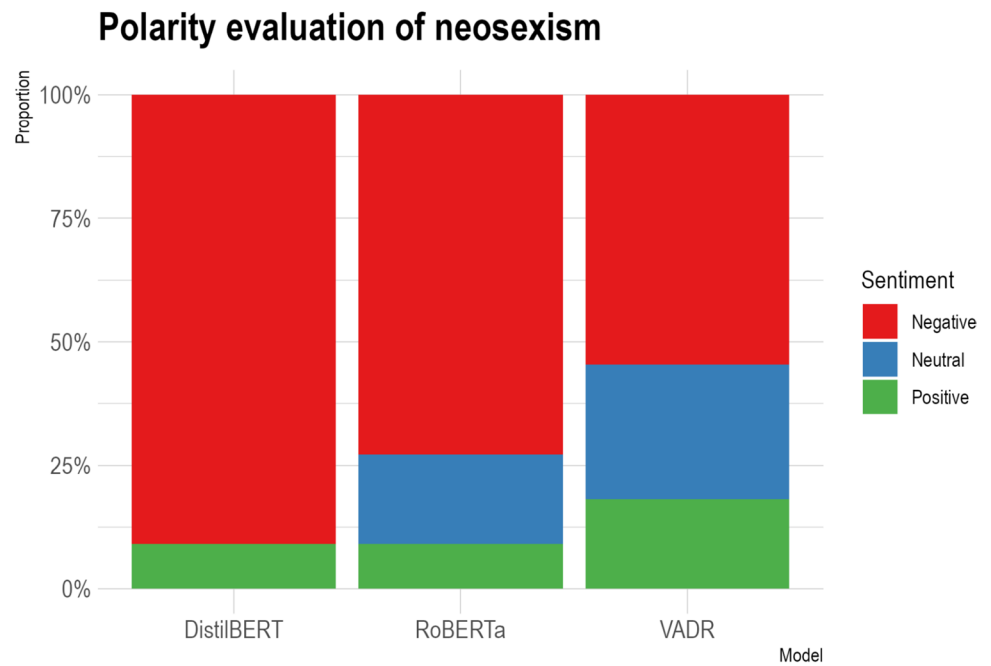
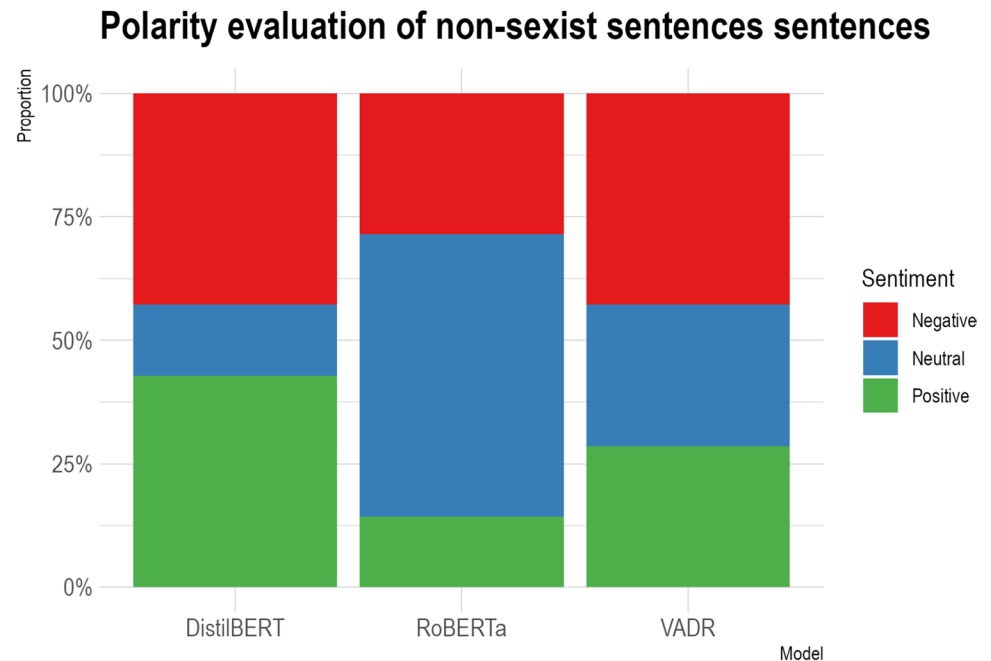


Fig. 5 Polarity evaluation of non-sexist sentences per model

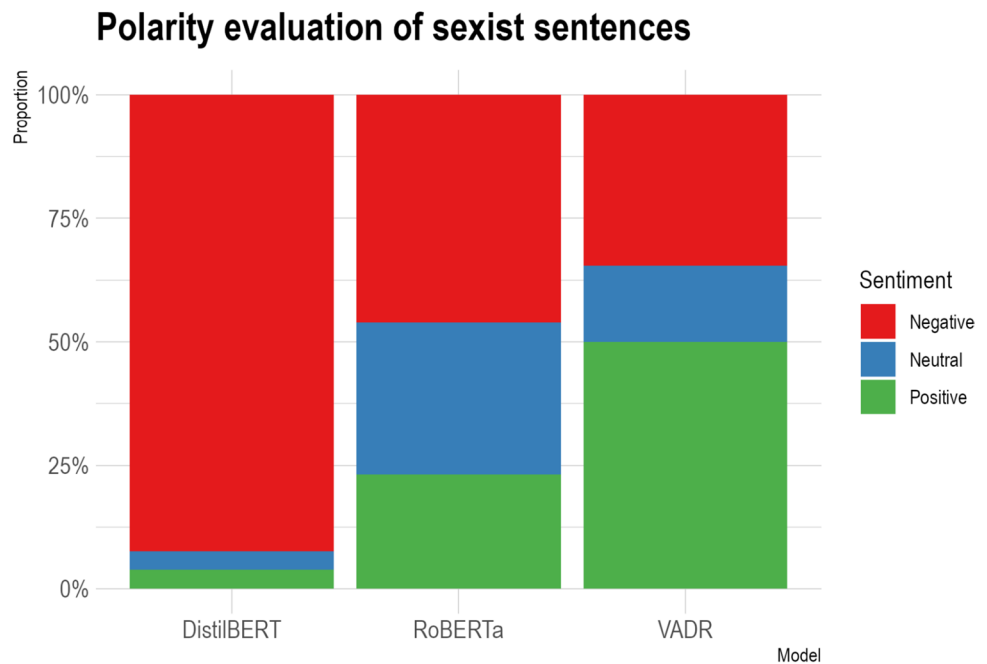


As the figure shows, DistilBERT was the only model to classify over 50% of sexist sentences overall as negative, with around 90% of the sentences. RoBERTa had almost 50% of the sentences as negative, but it also has a high percentage of neutral ones. Finally, VADER considered 50% of the sentences to be positive, which shows it is the model with the least awareness.

5 Discussion

Through a general look at the results obtained by simulating a normal use of Sentiment Analysis on sexist language, we can see there are answers to both Research Questions posed at the beginning. For the first one, we can see how

Fig. 6 General polarity evaluation of the 26 sexist sentences in the list



VADER did the worst job at classifying sexist expressions as negative, while DistilBERT considered most sexist sentences to be negative and RoBERTa had more negative sentences than VADER, but also classified more sentences as neutral. This hints towards Transformer-based models being better suited to detecting sexism in language than lexicon-based ones. However, it also raises concerns regarding the training of models, as RoBERTa did not consider over half of the sexist sentences to be negative, showing a lesser polarization around gender overall by classifying many sentences as neutral.

Looking at the results before and after the elimination of non-sexist sentences, however, shows how the likelihood of classification as positive or neutral for lexicon-based models is not related to the presence of sexism or lack of it in a sentence, and is based mainly on the superficial emotional appearance of the language employed (the form), rather than being related to the sexism of the message or the topics approached (the content). The Transformer-based models appear to have a better awareness of the presence of sexism due to them taking the context into account, but it seems that this awareness alone does not make them consider sexism to be negative, or they might not always include the possibility of sexism in their understanding of the context. This would make sense with the differences between DistilBERT and RoBERTa, with the former classifying most sexist expressions as negative while the latter did not appear to consider it as either negative or positive, or not as negative as DistilBERT seems to.

Regarding the second research question, the results show a clear difference between the way benevolent sexism is

evaluated in comparison to hostile sexism and neosexism. Although DistilBERT classified most sentences in every category as negative, RoBERTa and VADER had most of the benevolent sexism sentences as either positive or neutral, and RoBERTa did not consider any of them to be negative. This fits the way benevolent sexism works generally, as it is harder to detect due to the fact that it seems to be supportive of women on the surface and that it appeals to traditional standards that are naturalized and not seen as necessarily sexist (Glick and Fiske 1996). This points to the likelihood of many SA models having a harder time understanding benevolent sexism as negative, since even a state-of-the-art Transformer-based model had trouble evaluating these sentences. In order to be able to detect sexism with this technology, benevolent sexism would have to be accounted for by researchers and taken into account when creating the training sets, making sure that sexist expressions of any kind are understood as negative.

Given these results, we can argue that many Sentiment Analysis models are not aware of sexist language as a specific mode of expression with its own meanings and implications, and that this awareness requires specialized training in order to include it as part of their framework. This also reveals the unawareness of sexism or the indifference towards it as a hegemonic view within the world of AI development, acting as a default position. Of course, Sentiment Analysis models themselves cannot understand what sexism is at a structural, historical level, but at best learn to associate sexist words and expressions to negativity if the developers of the models have a perspective that makes this relevant. This, however, is at odds with the feminist

perspective, which acknowledges the historical development of women's oppression and the struggle against it, and considers this history as an important element to take into account. While SA models can then be used as tools to detect sexism in language if trained properly and this is a valuable tool to think of for feminist research, it also shows it to be a more superficial view of feminism that needs to be accompanied by a historical, structural perspective.

While useful for some studies, the way of treating sexism that Sentiment Analysis can implement comes from a framework of technological rationality (Houkes 2009; Marcuse 1941), which considers structural or social problems to be technical ones, thus solvable through the use of technology. This exemplifies the separation from the sociality and historical background of our environments that the scientific field finds itself in, and that it can in turn reproduce through the use of technology developed under such a context (Adorno 1998; Mumford 1970). The adoption of tools based on Machine Learning or other AI technologies in research and their further implementation in other practical work (Bhangdia et al. 2021), when understood as tools that can be used by themselves rather than needing human input and interpretation, can perpetuate this separation and result in a naturalization of an understanding of sexism that is, at best, considering that sexism is the presence of sexist sentences or behaviors. This erases the structural and material reality of sexism and its bigger implications, and reduces its scale to an intersubjective one rather than a political and economic reality.

The way the use of AI is spreading into many different scientific fields and becoming more relevant also signals another step in the way society understands knowledge. Cybernetic or mathematical information becomes more relevant or superior to other forms of obtaining information, which are relegated to past forms of research (Pasquinelli 2015). This results in an increase in the process of depoliticization that has been present for a long time in engineering and other fields (Cech and Sherick 2015). Depoliticization also becomes a secondary issue for many, since current developments of AI technologies work towards making the process of extraction and presentation of information as efficient and productive as possible, focusing on a standpoint of productivity and speed in order to generate benefits (Engster and Moore 2020). Challenging the hegemonic frameworks in science requires a conscious effort, as they work in a cycle that motivates the scientific field to keep on developing itself in the same direction.

This situation makes the issue of bringing epistemology to the forefront more pressing, but we must also question how to make changes happen in the technological field. It is not simply an exercise of using these technologies for good that would be required from professionals who wanted to implement them in their practice, but a transformation that

goes beyond the technical aspects and happens in the social sphere (Brassier 2014). Trying to solve the issue of depoliticization in science or of the reproduction of sexism by creating new ways to do research and simply bringing other forms of knowledge to light would be falling into technological rationality again, as it would just be yet another way of trying to solve a social issue through scientific advancement. The objective of the feminist epistemological critique here has to imply going beyond the scientific solutions and understanding that a substantial change in science can only be done through a transformation of the material conditions and structures that determine women's oppression in the first place.

While the findings of this study have great implications for the use of Sentiment Analysis models and other automated methods in the study of the human mind, there are also some limitations to the work presented. First of all, the amount of different Sentiment Analysis models available is quite substantial and rapidly growing, and this study does not implement all of the different models there are. Using several models and employing two different methods, lexicon-based analysis and transformers architecture, this study intends to reflect the state of the field. However, it cannot cover systems that might be implementing sexism-specific analysis in the future, nor those that have been made only with the aim to detect sexism. Secondly, this study limits the database employed for its analysis to two different sexism inventories, therefore not representing the whole scope of sexism and the different modalities it presents, especially when it comes to those that are more complex, such as misogynoir or transmisogyny. This could be an avenue of study for the future, and also serve to approach the critique of Sentiment Analysis models not only from a gender perspective but also a queer or decolonial one. Finally, this study has limited itself to studying sentences written in the English language. This derives from a limitation that the Sentiment Analysis models also have, as they are much more competent in this language due to the origin of most of the development we see in these technologies. As these systems rapidly evolve, the critique of them will have to be expanded and updated to cover the new grounds that Sentiment Analysis and other automated methods open to us.

6 Conclusion

In this study, we have intended to elaborate on the feminist epistemological critique of technology by extending this critique to the field of Sentiment Analysis. Through our findings, we have seen that SA models diverge in their ability to detect sexist expressions and sentences. Newer, transformers-based models can detect most sexist expressions as negative, making them mostly able to detect different forms of

sexism. However, detecting sexist sentences does not make Sentiment Analysis feminist. The deeper understanding of a feminist epistemological critique of these systems points towards social transformation as key to the creation of technologies that can be useful to challenge oppressive structures, rather than a reformulation of technological devices without transforming social structures.

While the ability to detect sexism that Sentiment Analysis models have can be an indicator of sexist biases in their development, a deeper angle of study is needed to truly understand the connection between these technologies and gender oppression. By discussing not only the way different models analyze different forms of sexism, but also the way these systems work, we can see how SA can only be used to grasp the more superficial levels of gender oppression through its manifestations in language. In this way, even if these technologies can detect sexism, they still contribute to the hiding of historical and material developments that have resulted in the current state of the oppression of women and other marginalized communities, ultimately perpetuating the sexist status quo.

While Sentiment Analysis and other automated methodologies offer a new way to conduct research, streamlining information and pushing us forward into the future, it is possible that these technologies might prove themselves harmful when trying to approach the human mind. Working from a critical feminist perspective, Sentiment Analysis models can be considered to be dangerous as they can only tackle superficial issues regarding our current societal structures and, furthermore, push the social sciences away from the creation of human knowledge about human emotions. Thus, from this point of view, it might be more commendable to develop more human, comprehensive ways to study emotions in the different social science disciplines rather than give in to the use of technologies whose shiny future could actually be just a loss of direction.

Appendix 1: Ambivalent Sexism Inventory

Hostile sexism

1. Women exaggerate problems they have at work.
2. Women are too easily offended.
3. Most women interpret innocent remarks as being sexist.
4. When women lose to men in a fair competition, they typically complain about being discriminated against.
5. Many women are actually seeking special favors, such as hiring policies that favor them over men, under the guise of asking for "equality".
6. Feminists are making entirely reasonable demands of men.*

7. Feminists not seeking for women to have more power than men.*
8. Women seek power by getting control over men.
9. There are actually very few women who get a kick out of teasing men by seeming sexually available and then refusing male advances.*
10. Once a woman gets a man to commit to her, she usually tries to put him on a tight leash.
11. Most women fail to appreciate all that men do for them.

Benevolent sexism

Protective paternalism

12. A good woman should be set on a pedestal by her man.
13. Women should be cherished and protected by men.
14. Men should be willing to sacrifice their own well being to provide financially for the women in their lives.
15. In a disaster, women need not be rescued first.*

Complementary gender differentiation

16. Women, compared to men, tend to have a superior moral sensibility.
17. Many women have a quality of purity that few men possess.
18. Women, as compared to men, tend to have a more refined sense of culture and good taste.

Heterosexual intimacy

19. Every man ought to have a woman he adores.
20. Men are complete without women.*
21. No matter how accomplished he is, a man is not truly complete as a person unless he has the love of a woman.
22. People are often truly happy in life without being romantically involved with a member of the other sex.*

**Items are reverse coded.*

Appendix 2: Neosexism scale or neosexism inventory

1. Discrimination against women in the labor force is no longer a problem in Canada.
2. I consider the present employment system to be fair to women.*
3. Women shouldn't push themselves where they are not wanted.
4. Women will make more progress by being patient and not pushing too hard for change.

5. It is difficult to work for a female boss.
6. Women's requests in terms of equality between the sexes are simply exaggerated.
7. Over the past few years, women have gotten more from government than they deserve.
8. Universities are wrong to admit women in costly programs such as medicine, when in fact, a large number will leave their jobs after a few years to raise their children.
9. In order not to appear sexist, many men are inclined to overcompensate women.
10. Due to social pressures, firms frequently have to hire underqualified women.
11. In a fair employment system, men and women would be considered equal.*

*Reverse scored item

Acknowledgements To the reviewers, who helped improve this paper beyond the scope we had imagined in the beginning.

Author contributions D.D.F. wrote the main manuscript, did the technical analysis and prepared the tables and figures. P.M.B. revised the manuscript and contributed to the references as well as to the theoretical framework section.

Funding Open Access funding provided thanks to the CRUE-CSIC agreement with Springer Nature.

Data availability The data employed in this study are available under a GNU General Public v3.0 license at <https://github.com/Dariodofa/Sexism-in-Sentiment-Analysis-data>.

Declarations

Conflict of interest The authors declare no competing interests.

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